

TurtleStitch in the classroom
What to do if USB ports are blocked?
Summary presentation

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Agenda

- 1) The problem: limitations around getting .DST files to the embroidery machines
e.g. because USB ports are blocked
- 2) Options to explore
- 3) Next steps

The problem

Increasingly, teachers using TurtleStitch in schools are confronted with new limitations

- 1) USB ports can no longer be used for USB sticks as storage device (AMP)
- 2) Files can only be put in the Downloads folder (H)
- 3) No executables can be run, not even from the Downloads folder (AP)
- 4) Only one prescribed Wifi access point can be accessed (PmA)
- 5) No Bluetooth can be used for file sharing (P)
- 6) Embroidery machines with Wifi access are not allowed in the network (PM)
- 7) In e.g. China, access to internet is blocked in some cases (S)
- 8) Externals / teachers cannot access the school's Wifi (MP)
- 9) Cannot send mail with .DST attachments (even when zipped and encrypted)

A	Andrel
B	Bice
H	Hans
M	Mararet
P	Pauline
S	Simon

Lower case =
some restrictions
apply

Questions:

- a) Are there more obstacles observed?
- b) Who encounters which obstacle?
- c) Does downloading to micro:bit via the browser still work for all? It uses WebUSB
- d) Same question for Arduino?

Direction to further explore

- Use a private laptop to write to the USB stick
- Let the private laptop get the .XML and .DST file via
 - Email
 - Now possible if participants have access to email
 - Via a server, TurtleStitch server or other
 - Now possible for .XML if on-line access to TurtleStitch is not blocked
 - For .DST to be explored how to do this as well
- We can automate parts of the process
 - The only usable thing for automation on the participant laptop is a browser that is running JavaScript program

In cases there are other options

- See e.g. DropStitch
 - Can be found via <https://dropstitch.design.blog/>
 - Created by the brother of Margaret Low
 - It runs on a Raspberry Pi Zero W
 - It can be plugged into the USB port of the embroidery machine
 - It connects to the laptop via Bluetooth
 - It could be used on the teacher's laptop when Bluetooth is allowed and then the participants could copy .DST files to the teacher's laptop via a shared drive
- There may also be options via bitlocker encrypted USB sticks to get the file to the private laptop
 - To be further explored

Next steps

- Simon and Hans will explore exchanging .DST file via servers
- And see how much can be automated with JavaScript in a browser
- It will most likely take several months before we have results.